CCC-C-438b
August 19, 1965
SUPERSEDING
Fed. Spec. CCC-C-438a
October 12, 1961
(See section 6)

FEDERAL SPECIFICATION

CLOTH, BUCKRAM, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cotton buckram.

1.2 Classification.

1.2.1 *Types and classes.* The finished cloth shall be of the following types and classes, as specified (see 6.2).

Type I—Resin finish.
- Class 1—5.7 ounces per square yard (fine texture).
- Class 2—5.7 ounces per square yard (medium texture).

Type II—Nonchlorine retentive resin finish.
- Class 1—4.5 ounces per square yard.

Type III—Starch finish.
- Class 1—3.8 ounces per square yard.
- Class 2—12.0 ounces per square yard.

Type IV—Starch finish, double texture (2 plies of type III, class 2).
- Class 1—24.0 ounces per square yard.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-T-191—Textile Test Methods.

PPP-P-51—Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

FSC 8305

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Application for copies should be addressed to the Federal Trade Commission, Washington, D. C., 20580.)

3. REQUIREMENTS

3.1 **Standard sample.** The dyed and finished cloth shall match the standard sample for shade; and the natural, bleached or dyed and finished cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 **Material.**

3.2.1 *Yarn.* The yarn shall be made from cotton which has been cleaned, carded, drawn and spun into singles yarn for both warp and filling.

3.3 **Color.** The color shall be natural, bleached or dyed as specified (see 3.3.2 and 6.2). When natural is specified, the cloth shall be fabricated from unbleached yarn. The use of reused or dyed fiber will be allowed, subject to the colorfastness tests specified in 3.3.2. The use of sulfur dyes or dyes containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more free or sulfide sulfur than the standard sample when tested as specified in 4.4. For civil agencies, when no standard sample is available, there shall be a minimum amount of staining of the test paper when tested as specified in 4.4.

3.3.1 *Matching.* The color shall match the standard sample (see 6.3), under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin, and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.*

3.3.2.1 *Natural cloth.* When dyed fibers are utilized (see 3.3), the finished cloth shall show fastness to laundering, perspiration, and wet drycleaning equal to or better than the standard sample (see 6.3). When the standard sample is not established for colorfastness, the "natural" and finished cloth (when composed of dyed fibers) shall show "good" colorfastness to laundering, perspiration, and wet drycleaning when tested as specified in 4.4.

3.3.2.2 *Dyed cloth.* The dyed and finished cloth shall show fastness to laundering, crocking, perspiration and wet drycleaning equal to or better than the standard sample (see 6.3). When the standard sample is not established for colorfastness, the dyed and finished cloth shall show "good" colorfastness to laundering, perspiration, and wet drycleaning and shall show a Munsell Value for crocking not lower than 8.5. Tests shall be as specified in 4.4.

3.4 **Physical requirements.** The finished cloth shall conform to the requirements in table I when tested as specified in 4.4.

3.4.1 *Weave.* The weave shall be plain, except that type III, class 2 and type IV, class 1 shall be a herringbone twill.

3.4.2 *Width.* The width of the cloth shall be as specified (see 6.2), and shall be the minimum acceptable width inclusive of selvages.

3.5 **Finish.**

3.5.1 *Type I.* The type I cloth shall have

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TABLE I. *Physical requirements*

Type	Class	Weight per sq. yd., minimum	Yarns per inch, minimum		Breaking strength (pounds), minimum		Stiffness (inch-pounds), minimum		Shrinkage (percent), maximum		Non-fibrous materials, maximum
		Ounces	*W*	*F*	*W*	*F*	*W*	*F*	*W*	*F*	*Percent*
I	1	5.7	52	44	63	54	0.020	0.010	3.0	2.0	25
	2	5.7	44	22	80	54	0.034	0.018	3.0	3.0	10
II	1	4.5	50	42	60	50	—	—	1.0	1.0	60
III	1	3.8	46	42	50	40	—	—	—	—	35
	2	12.0	48	20	140	125	—	—	—	—	50
IV[1]	1	24.0	—	—	140	125	0.740	0.570	—	—	—

[1] Two plies of type III, class 2 cloth.

a resin finish and the total nonfibrous materials, including resins, plasticizers and other substances, shall not exceed the amounts specified in table I.

3.5.2 *Type II.* Type II cloth shall have a nonchlorine retentive resin type finish and the material shall show no discoloration or loss of no more than 10 percent of the original breaking strength of the cloth when tested as specified in 4.4. The total nonfibrous materials, including resins, plasticizers and other substances shall not exceed the amounts specified in table I.

3.5.3 *Type III.* Type III, class 1 cloth shall have a starch swing finish. Determination of the swing finish shall be made as specified in 4.4. Type III, class 2 cloth shall have a starch finish. The starch and protein content, including chloroform-soluble and water-soluble material for type III cloth, shall not exceed the amounts specified in table I when tested as specified in 4.4.

3.5.4 *Type IV.* Type IV cloth shall be made of two plies of type III, class 2 cloth. The two plies shall be joined by laminating the two plies together.

3.5.4.1 *Laminating material.* The laminating material shall be of a synthetic resin type, unaffected by wet drycleaning. There shall be no objectionable odor caused by the laminating materials.

3.5.4.1.1 *Resistance to wet drycleaning.* The laminated cloth shall show no evidence of delamination when subjected to three cycles of wet drycleaning when tested as specified in 4.4.

3.6 Shrinkage (types I and II). The finished cloth for types I and II shall conform to the shrinkage requirements in table I when tested as specified in 4.4. The preshrinking processes used shall not be identified by name or trademark, either on the cloth, ticket, or package.

3.7 Length and put-up. Unless otherwise specified, the cloth shall be furnished in continuous lengths each not less than 40 yards. Each length shall be put up in rolls as specified in PPP-P-51.

3.8 Fiber identification. Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.9 Workmanship. The finished cloth shall conform to the quality established by this specification. The points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection require-

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ments as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of MIL-STD-105 and 4.2.1 through 4.2.4.

4.2.1 *Yard-by-yard examination.* The required yardage of each roll shall be examined on the face side and all defects as defined in section 1 of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1, with the following exceptions:

Heavy places (glossary No. 2) shall not be scored.

Only those slubs and knots which exceed the maximum acceptable limits shown for each in figure 1 shall be scored. None of the 30 contiguous linear yards (increments of 1 yard on the measuring device of the inspection machine) selected for examination from any one roll within the sample, shall be penalized more than four points. The sample unit for this examination shall be one linear yard. The sample size shall be 750 linear yards, regardless of lot size. The sample yardage shall be examined from 25 rolls taken from 25 containers. An approximately equal number of yards shall be examined in each roll of the sample. The lot shall be unacceptable if the points per hundred square yards exceed the following:

Type I, class 1 —35.0 points
Type I, class 2 —30.0 points
Types II, III, IV—40.0 points

Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of the material (inches)}} = \text{points per 100 square yards}$$

4.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension —one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension —two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension —three points
For defects exceeding 9 inches in any dimension —four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Objectionable odor.
Width less than specified.
Poor dye penetration, mottled, streaky, or cloudy.

4.2.2 *Examination for length.*

4.2.2.1 *Individual rolls.* During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.2.3 *Examination for shade.* During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.4 *Examination for identification of preshrinkage process and noncompliance with the Textile Fiber Products Identification Act.* During the yard-by-yard exami-

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nation, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-51 to determine that packaging, packing, and marking requirements comply with section 5.

4.4 Testing of the end item. The methods of testing specified in CCC-T-191 wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be as follows:

a. Four yards, full width of the finished cloth, for type II, class 1.
b. Three yards, full width of the finished cloth, for type I, classes 1 and 2.
c. One and half yards, full width of the finished cloth, for types III and IV.

The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final result.

TABLE II. *Test methods*

Characteristic	Requirement reference	Test method
Weight	Table I, 3.4	5041
Yarns per inch	Table I, 3.4	5050
Breaking strength	Table I, 3.4	5100
Weave	3.4.1	Visual[1]
Swing finish (type III, class 1)	3.5.3	4.4.1[1]
Chlorine degradation (type II)	3.5.2	4.4.3
Discoloration (type II)	3.5.2	4.4.4
Stiffness	Table I, 3.4	5202
Colorfastness to:		
Laundering	3.3.2	5610
Wet drycleaning	3.3.2	5622
Crocking	3.3.2	5651
Perspiration	3.3.2	5680
Sulfur content	3.3	4.4.2[1]
Nonfibrous materials:		
Type I	Table I, 3.5.1	[2]
Type II	Table I, 3.5.2	[2]
Type III	Table I, 3.5.3	2611[2]
Shrinkage in laundering (types I and II)	Table I, 3.6	5550
Resistance to:		
Wet drycleaning for type IV (3 cycles)	3.5.4.1.1	5622[3]

[1] One determination shall be made from each sample unit and the results reported as "pass" or "fail".

[2] The supplier shall submit a certificate of compliance for this characteristic. When required, the certificate shall be accompanied by actual test, inspection or other verifiable quality data.

[3] The specimen shall be examined for delamination. If the material exhibits delamination, it shall be cause for rejection.

4.4.1 *Swing finish.* A piece of type III, class 1 cloth 10 by 10 inches shall be folded on a 45° bias without creasing. Hold the ends of the fold with both hands and pull on both ends of the cloth. Cloth having a swing finish will stretch or buckle.

4.4.2 *Test for sulfur content.*

4.4.2.1 *Solution (acid stannous chloride).* Dissolve 100 g. of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water.

4.4.2.2 *Procedure.* One to 1.5 g. of finished cloth, cut into small pieces shall be used for the test. Place the sample in a test tube of suitable size and cover with stannous chloride solution. Place a filter paper which has been moistened with one drop of 5.0 per-

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cent lead acetate solution (made by dissolving 5 g. of reagent grade lead acetate to 100 ml. of distilled water; add a drop of glacial acetic acid to clear up if cloudy) over the mouth of the test tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds, as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

4.4.2.3 *Evaluation.* The stained filter paper shall then be compared with that obtained from the standard. If the stained area is equal to or lighter than the standard in the degree of staining, the sample shall be considered satisfactory. If the degree of staining by the sample is in excess of the standard, the sample shall be considered unsatisfactory. For civil agencies, when no standard sample is available, there shall be a minimum of staining on the paper.

4.4.3 *Chlorine degradation (type II).* A piece of cloth, not less than 20 inches long and full width, shall be laundered in accordance with method 5550 of CCC-T-191 and shall then be treated for one hour with 0.7 percent available chlorine solution. The piece shall be rinsed in a sour solution and washed thoroughly until a control piece shows no trace of chlorine. The piece shall be air-dried and then conditioned for 4 hours in an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of 75° ± 5°F. Eight pairs of specimens, four warpwise and four fillingwise, with each pair having the same yarns, shall be taken for tests. Four warp specimens and four filling specimens shall be tested in a pendulum-type machine in accordance with method 5100 of CCC-T-191. The remaining warp and filling specimens shall be oven dried at 105° to 110°C. for two hours and tested in the same manner as the first eight specimens. The difference in the average breaking strength of the respective warp and filling samples shall be computed and the percentage of loss determined. If a specimen slips or breaks in or at the edge of the jaws, the result shall be discarded and another pair of specimens taken, both having identical warp and filling yarns and the test repeated on this pair of specimens.

4.4.4 *Discoloration test (type II).* A 4 by 4 inch piece of the chlorinated and conditioned sample of cloth from 4.4.3, with a piece of the unchlorinated cloth shall be oven dried at a temperature of 105° to 110° C. for two hours. Both pieces shall be removed from the oven and allowed to cool at room temperature. The chlorinated piece shall then be compared with the control cloth for discoloration. Any discoloration shall be cause for rejection.

5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A and C.* The cloth, put up as specified, shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-51.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.2.1. *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP-P-51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments, shall be marked in accordance with the requirements of PPP-P-51.

6. NOTES

6.1 Intended use. Types I, II, and III cloth are used as interlining for clothing. Type II cloth is suitable for lining shirt collars, and type IV cloth (2 ply) is used when

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a high degree of stiffness is desired, as in the standup collar for cadet uniforms.

6.1.1 *Type I, class* 1. Used in:

Air Force—Man's summer coat.
Navy—Men's and women's uniforms.
—Men's and women's overcoats (except pea coats).
Marine Corps—Coats and belts.

6.1.2 *Type I, class* 2. Used in:

Air Force—Wool serge coat.
Army—Man's and woman's coat.
—Man's overcoat.
Marine Corps—Man's overcoat.
—Man's coat.
—Woman's coat.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of cloth required (see 1.2).
(*c*) Color of cloth required (see 3.3).
(*d*) Width of cloth required (see 3.4.2).
(*e*) When length is other than specified (see 3.7).
(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives as indicated in PPP-P-51, may be exercised in the preparation for delivery of this commodity.

6.3 **Standard sample.** For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Comparison of types and classes as pertains to CCC-C-438 and MIL-C-340 is as follows:

CCC-C-438a	*CCC-C-438b*
Type I (classes 1, 2, 3)	Deleted
Type II, class 1	Type III, class 1
Type II, class 2	Type II, class 1
Type II, class 3	Type III, class 2
Type II, class 4	Deleted
Type III	Type IV

MIL-C-340a	*CCC-C-438b*
Class 1	Type I, class 1
Class 2	Type I, class 2

6.5 **Supersession data.** This specification includes the requirements of MIL-C-340, dated October 24, 1962.

6.6 **Transportation description.** Transportation description and minimum weights applicable to this commodity are:

Rail:

Buckram, cotton.
Carload minimum weight—Any quantity.

Motor:

Buckram, cotton.
Truckload minimum weight—Any quantity subject to Rule 115, National Motor Freight Classification.

6.7 Level B packaging and packing shall be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

MILITARY CUSTODIANS:

Army—GL
Navy—SA
Air Force—11

Review activities:

Army—MD
Navy—SA, MC
Air Force—11, 67

User activities:

Army—MO, MR, EL
DSA—DM

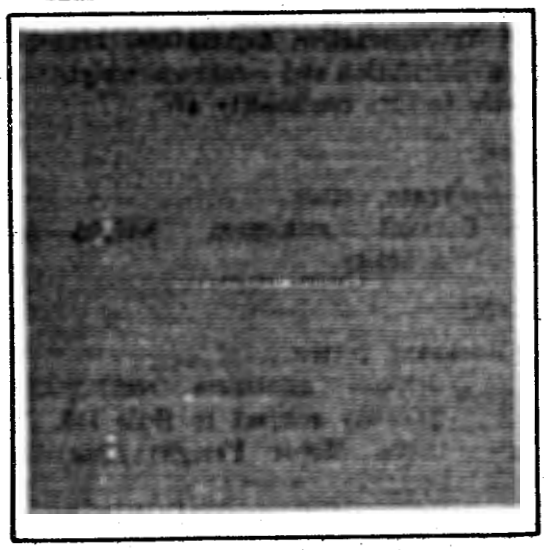
Preparing activity:

Army—GL

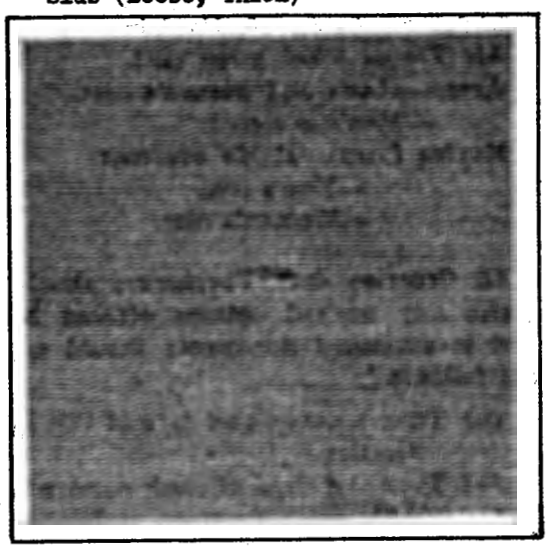
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Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)

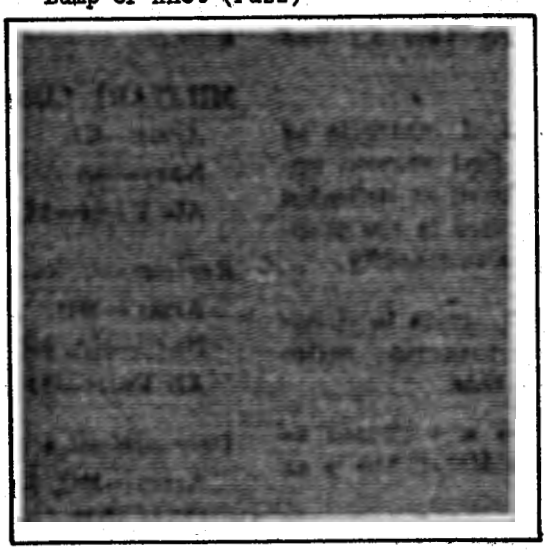


FIGURE 1. Limits of acceptability

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